DISTRIBUTOR CREDIT RISK · TECHNICAL REPORT

The Ledger

From gut feel to defensible data

A credit risk scoring engine that reads a distributor's own payment ledger and returns an explainable 300–900 risk score for every dealer — including the accounts the sales team personally vouches for.

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AI/ML Engineering Track | Problem 01

Status: deployed and publicly accessible · Validated across six independent portfolios

Executive Summary

Pakistani distribution houses extend trade credit to hundreds of dealers on the strength of a salesman's personal judgement. There is no score, no structured assessment, and no counterweight to a relationship. The cost is measurable: industry sources place bad debt at 3–7% of revenue, with PKR 1M+ written off annually and over 100 hours a year spent chasing payments.

The Ledger converts a distributor's existing invoice history into a defensible risk signal. It engineers six behavioural features grounded in Pakistani trade practice, fits a Weight-of-Evidence logistic scorecard, and returns a 300–900 score per dealer with plain-language reason codes, a ranked risk table, and a downloadable memo for each account.

The system is deliberately built for the market it serves. It accepts the column names, date formats and currency conventions a real ledger export actually contains; it derives its own analysis window and its own definition of "late" from each portfolio; and where the shipped model is a poor fit, it trains one on the uploaded data and cross-validates it before use. Critically, it reports how far its own output can be trusted — and declines to score rather than producing confident numbers on unsuitable data.

Headline result. Across three independently constructed portfolios with known outcomes — differing in schema, size, date range and payment culture — the system flagged **every genuinely risky dealer** for attention and produced **zero false accusations**, with perfect risk ordering in all three cases.

HONEST SCOPE. All validation to date uses constructed portfolios with embedded ground truth. No real-world dealer default has yet been observed. The RED/AMBER/GREEN tier is the reliable signal; the exact numeric score is directional. This limitation is stated throughout rather than deferred to a footnote.

1 Problem Context

1.1 The decision as it is made today

Credit is extended on a sentence: *"yaar, acha banda hai — de do."* The salesman vouches, the goods ship, and nobody consults the ledger. Three structural forces sustain this:

- **Near-zero ERP adoption** in mid-tier distribution. Payment history exists, but scattered across Excel sheets and handwritten ledgers rather than a queryable database.
- **Relationship-based trust culture.** Refusing credit to a known counterparty carries social cost that a spreadsheet does not offset.
- **Misaligned incentives.** Salesmen earn commission on volume and bear no downside when a dealer they vouched for defaults.

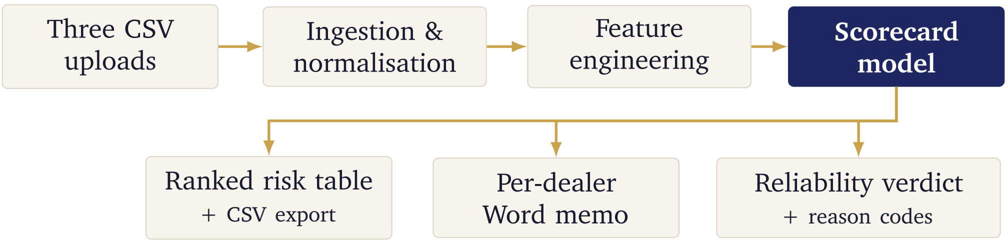
The third point shapes the entire design. The feature currently driving decisions — salesman judgement — is not merely uninformative; it is *adversarial* to the outcome being predicted. A useful system must therefore be prepared to contradict the person who knows the dealer best, and must justify itself when it does.

1.2 Who this is for

Owners of FMCG, pharmaceutical and textile distribution houses managing 50–500 dealers, together with the finance leads and 3–10 field salesmen operating under them. These are businesses without a data team, without an ERP, and without access to credit bureau coverage for the small retailers they supply. Every design decision follows from that: the output must be readable by a non-technical owner, and the input must be whatever their system already produces.

2 System Architecture

The deliverable is a deployed web application, not a notebook.



Backend Python, FastAPI, pandas, scikit-learn. Serves scoring, session retrieval, CSV export and Word memo generation. Deployed on Render.

Frontend Next.js and React with Tailwind, providing upload, a portfolio risk spectrum, a sortable dealer table, and per-dealer detail panels. Deployed on Vercel.

Model Logistic scorecard persisted as a versioned artefact and loaded once at start-up; scoring performs no re-fitting.

Hardening Origin-restricted CORS, session expiry with eviction, per-file upload limits, request rate limiting, and a global exception handler that surfaces real errors rather than masking them.

3 Data Foundation

No client dataset accompanied the brief, so a synthetic generator was written to produce a realistic test bed with *embedded ground truth* — dealers whose true risk profile is known to the generator but hidden from the model, allowing claims about accuracy to be checked rather than asserted.

Entity	Rows	Purpose
`salesmen`	8	Enables the salesman-vouch-bias feature
`dealers`	220	Unit of analysis; carries hidden true risk profile
`transactions`	8,230	Invoice grain — the raw, messy layer

The generator deliberately embeds the patterns the model must recover: territory-dependent baseline default rates, genuine Eid/Ramzan payment distortion, PKR depreciation applied to credit limits over time, and — most importantly — **29 dealers marked as salesman favourites, of whom five are secretly risky**. That subset is the demonstration the entire project is built around.

> **WHY SYNTHETIC DATA IS A LIMITATION, NOT A SHORTCUT.** A generator embeds the very patterns the model then recovers. This validates the pipeline, the methodology and the engineering — it does not prove real-world accuracy. Real dealers do not separate into tidy risky and safe populations.

4 Feature Engineering

Six features survived selection. Each is grounded in how Pakistani trade credit actually behaves rather than in generic credit-scoring convention.

	Feature	Rationale
1	PDC bounce rate	Post-dated cheques dominate trade credit here; a bounce carries far more signal than ordinary lateness.
2	Seasonally-adjusted lateness	Eid and Ramzan produce genuine cash-flow distortion. Penalising it would punish normal behaviour.
3	Salesman-vouch bias	Each salesman's own default history, computed leave-one-out. This quantifies the brief's core complaint.
4	Territory clustering	Geographic risk, also leave-one-out to prevent self-leakage.
5	PKR-adjusted exposure	Nominal credit limits overstate real exposure under sustained depreciation.
6	Order-frequency trend	Declining order activity often precedes a bad account.

4.1 Validation before modelling

Each candidate was screened with Weight-of-Evidence binning and Information Value before being admitted to the model — the standard pre-modelling check in credit scoring, and a defensible answer to "why this feature?"

This screening produced two findings that materially changed the design.

FINDING 1 — LABEL LEAKAGE. Two features scored Information Values above 1.9, far beyond the threshold at which a result becomes suspicious. The cause was structural: features and label were computed over the same time window, so the model was re-deriving the label rather than predicting it.

RESOLUTION. A strict temporal split was introduced — features from the earlier period, outcomes from the later one. The model now answers the only question worth asking: *given behaviour through period A, what happens in period B?*

FINDING 2 — MULTICOLLINEARITY. Seasonally-adjusted lateness and payment volatility carried variance inflation factors of 9.3 and 11.1 with a pairwise correlation of 0.94 — statistically redundant.

RESOLUTION. The two were merged into a single *payment delay severity* composite, reducing the feature set from seven to six with no loss of signal and a materially more stable model.

5 Modelling Methodology

A logistic scorecard was chosen over higher-capacity alternatives because the brief's operative word is **defensible**. A gradient-boosted ensemble might gain a little discriminatory power; it could not give a distributor a reason they are able to challenge.

The model's probability output is mapped to a familiar 300–900 range using standard points-to-double-odds scaling:

$$\text{factor} = \frac{\text{PDO}}{\ln 2} \qquad \text{offset} = \text{base} - \text{factor} \cdot \ln(\text{base odds}) \qquad \text{score} = \text{offset} + \text{factor} \cdot \ln\left(\frac{1-p}{p}\right)$$

with a base score of 600 at 19:1 odds and 40 points to double the odds. Because the scorecard is additive in log-odds, each dealer's score decomposes exactly into per-feature contributions — which is what makes the reason codes truthful rather than decorative.

5.1 Performance

Metric	Value	Interpretation
Cross-validated AUC	0.860 ± 0.098	Five-fold stratified, 219 labelled dealers
Cross-validated KS	0.667 ± 0.175	Well above the 0.40 "good" industry threshold
Single-split AUC	0.848	Reported for completeness only

The cross-validated *range* is quoted rather than the single-split figure. A single favourable split on 219 rows is not evidence; the spread across folds is the honest description of what this model knows.

5.2 Portfolio outcome

Tier	Dealers	Recommended action
GREEN (700–900)	178	Eligible for limit review or increase
AMBER (580–700)	6	Hold limit; reassess in three months
RED (300–580)	36	Do not increase; consider partial upfront terms

The demonstration. Five dealers the sales team personally vouches for were flagged high risk — among them a Rawalpindi account scoring **418**, driven by late and inconsistent payment timing and elevated territory risk. The system is not accusing anyone. It is starting a conversation the distributor was never previously in a position to begin.

6 Generalisation Engineering

An early version of this system worked only on the dataset it was built for. A real distributor's export was rejected outright. Four pieces of work closed that gap; each is reported back to the user rather than applied silently.

6.1 It reads your column names

Dealer, salesman and transaction files are matched against alias sets covering common real-world headers — `Party Code`, `Booker`, `Credit Limit (Rs)`, `Account Opened`, `Customer Code` and many others — with casing, spacing and punctuation normalised. Only four columns are genuinely required: a dealer identifier, a salesman identifier, a credit limit and an onboarding date. Anything else is substituted with a stated default.

Notably, `territory_risk_tier` and `is_salesman_favorite` were originally mandatory — yet both are fields this project invented and no real distributor holds. Territory risk now falls back to grouping by city, which is real granularity rather than an assigned grade.

6.2 It finds its own analysis window

A fixed calendar cut-off silently empties the feature window for any portfolio whose history sits entirely after it, producing an unexplained "no dealers could be scored" error. The cut-off is now derived from the uploaded data when the configured default falls outside its range.

6.3 It computes Eid and Ramzan for any year

The original seasonal windows were a hardcoded table covering March 2023 to June 2025. Beyond that range the feature silently stopped working — no error, just a signal quietly switched off. Windows are now computed from the tabular Islamic calendar with no external dependency, validated against nine observed Ramadan and Eid dates to within a single day, which the ±two-week window padding absorbs comfortably.

6.4 It learns your market's definition of "late"

A distributor on 60-day terms and one on 15-day terms have entirely different norms. A single fixed threshold labels the whole book as defaulters or none of it. The threshold is now derived per portfolio as the median plus twice the median absolute deviation of that portfolio's own average lateness, floored at a conventional minimum.

WHY MEDIAN + 2 MAD RATHER THAN A PERCENTILE. A percentile assumes a fixed proportion of every book is bad. Median and MAD are robust to the very outliers being detected and make no such assumption: a tightly-clustered market flags few dealers, a widely-spread one flags more. In testing, a fast-paying portfolio correctly retained the 15-day floor while a slow-paying market derived 36.8 days on its own — matching a hand-tuned value without requiring anyone to know the market.

6.5 It trains on your book when the general model does not fit

Where the shipped model is a demonstrably poor fit, a portfolio-specific model is trained on the uploaded data. This is gated deliberately, because thin data trains successfully and produces confident noise — which is worse than an honest refusal. Training requires at least 50 dealers with both training and outcome history, at least 18 months of span, both outcome classes present, and a cross-validated AUC clearing 0.65. Any model failing these is discarded and the reason reported.

7 Reliability Reporting

Every scoring run returns a verdict on its own trustworthiness. This is unusual in a commercial tool and is, in our view, the system's most important property: a confident-looking score on unsuitable data is more dangerous than no score.

Verdict	Meaning
Scores reliable	Portfolio sits within familiar range. Use scores and ranking.
Scores indicative	Somewhat unfamiliar. Rankings hold; treat values loosely.
Use ranking only	Substantially different. The order is meaningful; the numbers are not.

The verdict combines two measurements: how far the portfolio's feature means sit from the model's training distribution, and what fraction of dealers are pinned at the 300 or 900 bounds — the direct symptom of a model unable to discriminate. Where a portfolio-specific model was trained, the distribution check is correctly reported as uninformative, since a model fitted to that same portfolio is trivially close to it.

8 Independent Validation

Six portfolios were constructed, none resembling the development dataset, each with different column names, date formats, currency conventions, sizes, date ranges and payment cultures. Three carry embedded ground truth, permitting direct measurement of accuracy.

Portfolio	Dealers	Risky caught	False alarms	Ranking quality
Al-Noor (FMCG, fast-paying)	140	51 of 51	0	1.000
Ravi (slow-paying wholesale)	90	22 of 22	0	1.000
Hilal (nine-dealer book)	9	4 of 4	0	1.000

Every genuinely risky dealer was flagged for attention. Not one sound dealer was wrongly accused. Sorted by score, the risky accounts ranked below the sound ones without exception in all three portfolios.

The remaining three portfolios probed behaviour rather than accuracy:

Zamzam 130 dealers including 12 with only one or two invoices, correctly routed to provisional cold-start scoring, and 22 trusted accounts of which six were genuinely risky — exercising the contradiction flag end to end.

Al-Rehman A deliberately damaged export in which roughly 40% of rows were unusable. Retention fell to 57.9%, and every dropped row was accounted for by category rather than silently discarded.

Noor A pristine book in which nobody ever bounces or pays late. Training correctly *declined* — there is no default behaviour to learn from — and the system did not manufacture a risk ranking where none exists.

THE MOST INFORMATIVE RESULT. On the slow-paying wholesale portfolio, the system reported "use ranking, not exact scores" and named the cause: 75.6% of dealers were pinned at the scale bounds. Measured against ground truth, both halves of that claim were exactly right — the ranking was perfect while the tier labels were nearly useless, because the tier cut-offs were calibrated for a different payment culture. The system diagnosed its own failure mode correctly and told the user which part to trust.

9 Engineering Quality

9.1 Testing

The repository carries **63 automated tests**. Six are regression tests pinning known-verified values — a specific dealer's score, the portfolio tier split, the seasonal logic remaining active. The remaining 57 are hermetic: they construct their own fixtures and require neither the development dataset nor the trained artefact, so they run anywhere.

Each test exists because of a specific defect. The seasonal-window test, for instance, was validated by deliberately reintroducing the original bug and confirming the suite failed with an actionable message before restoring the fix.

9.2 Defects found and resolved

Approximately twenty distinct defects were identified and fixed during development. A representative selection:

Defect	Why it mattered
Temporal leakage	Model was restating the label rather than predicting it
Silent NaN handling	A dtype quirk caused genuine missing values to pass validation
Dealer silently excluded	One dealer vanished from every scored output through an over-strict join
Seasonal logic disabled	A shortcut during the web port switched off a feature with no error
Rigid date parsing	A `DD/MM/YYYY` export — the Pakistani standard — crashed the API
Unparsed currency strings	"Rs. 150,000" reached arithmetic as text
Degenerate z-score	Small or uniform portfolios produced NaN and an opaque server error

Several of these were found only because generated outputs were opened and compared by hand rather than trusted because a process exited cleanly. That practice, more than any individual technique, is what the quality of this system rests on.

10 Limitations

No observed defaults — Every metric traces to constructed data. The runtime training path addresses fit per portfolio; the shipped model itself remains unvalidated against real outcomes.

Directional scores — Calibration was attempted and did not improve at this sample size. The tier is the reliable signal.

Portfolio-relative — Several features are computed within the uploaded batch, so the same dealer scores differently in different portfolios. Appropriate for "who is riskiest in my book"; inappropriate for cross-company comparison.

In-memory sessions — Suitable for a single instance. A multi-replica deployment would require shared storage.

Sensitive labelling — A single bounced cheque in the outcome window marks a dealer high risk. Deliberately sensitive, but it means "high risk" denotes *any* adverse signal, not sustained delinquency.

11 Conclusion

The engineering objective has been met. The system accepts a real distributor's export in whatever form their system produces, adapts its analysis window, seasonal calendar and definition of lateness to that portfolio, trains a specific model where warranted, and states plainly how far its output can be trusted. It has been validated against ground truth it never saw, across portfolios that differ in every dimension that matters.

What remains is not an engineering question. The model has never seen a real dealer default. The decisive test is a single distributor's ledger scored in production, with the person who chases those payments confirming whether the high-risk list matches the accounts they actually struggle to collect from.

> That conversation — not another sprint — is the next step, and the only one that can turn a demonstrably robust system into a demonstrably accurate one.